

Yissum Research Intelligence Report — Agriculture & Food — July 2026

2 paper(s) · Agriculture & Food · Monthly · July 2026 · Generated July 22, 2026

■ ■ There is no high-potential applicable research this period — only lower-scoring papers were found. Presenting the two best below.

Hebrew University researchers are advancing critical innovations in AgriTech and FoodTech, addressing key challenges in global food systems. One breakthrough offers a natural method to extend the shelf-life and visual appeal of premium beef products, reducing waste and enhancing consumer satisfaction. Another project deciphers a fundamental plant growth mechanism, paving the way for precision engineering of crops to significantly improve yields and resilience. These discoveries represent high-potential opportunities for investors seeking to impact the future of food production.

RESEARCHERS & RESEARCH SUBJECTS — BEST AVAILABLE

- **Oren Tirosh** — AgriTech, FoodTech
- **Naomi Ori** — AgriTech, Synthetic Biology, FoodTech

AGRICULTURE & FOOD — RESEARCH HIGHLIGHTS

1. Vitamin E supplementation moderates salt-induced oxidation status in beef. 23/50

"Innovative feed additive extends shelf-life and boosts visual appeal of premium salted beef."

Main researcher: Oren Tirosh · Oren.tirosh@mail.huji.ac.il.

Institute of Biochemistry, Food Science and Nutrition, The Robert H Smith Faculty of Agriculture, Food, and Environment, The Hebrew University of Jerusalem, Rehovot, Israel. Electronic address: Oren.tirosh@mail.huji.ac.il.

Published in Meat science · 2026-09-01

WHY NOW

Addressing significant industry losses from oxidation, this offers a natural, scalable solution to enhance meat quality. It targets the rapidly growing premium food market, boosting producer profitability and consumer satisfaction.

COMMERCIAL ANGLE

Development of specialized livestock feed additives or precision nutrition protocols to extend the shelf-life and visual appeal of premium salted beef products.

COMMERCIALISATION METRICS

Novelty

3/10

The research applies a well-known antioxidant (Vitamin E) to a known problem (lipid peroxidation) using standard dosage optimization; it lacks a novel mechanism, a proprietary delivery system, or a groundbreaking discovery.

Commercial Potential

6/10

The study identifies a clear, practical application for enhancing meat shelf-life and quality via a specific dosing regimen, but it lacks high-margin protectability as it utilizes a generic, well-known supplement (Vitamin E).

Market Size

6/10

The target market is the global beef industry, which is vast; however, this is an incremental improvement using a known additive (Vitamin E) rather than a disruptive platform technology.

Tech Readiness

6/10

The research has moved beyond basic proof-of-concept to in vivo optimization of dosing and duration in calves, demonstrating clear industrial applicability. However, it remains at a pilot/experimental stage and requires scale-up validation in commercial feedlots before being market-ready.

IP Strength

2/10

The research applies well-known nutritional concepts (Vitamin E supplementation) to a common problem, lacking a novel molecule, proprietary delivery system, or unique mechanism that would meet the non-obviousness threshold for strong patentability.

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2. A two-step auxin-GA cross talk regulates organ formation.

18/50

"Decoding plant hormone crosstalk to precisely engineer crop architecture and yields."

Main researcher: Naomi Ori · naomi.ori@mail.huji.ac.il

The Robert H. Smith Faculty of Agriculture, Food and Environment, The Hebrew University of Jerusalem, Rehovot 7610001, Israel.

Published in Development (Cambridge, England) · 2026-06-25

WHY NOW

This foundational discovery opens new avenues for synthetic biology and advanced breeding, promising to revolutionize crop productivity and resilience amidst global food security challenges.

COMMERCIAL ANGLE

Understanding this hormonal switch enables the precision engineering of plant architecture and crop yields through targeted metabolic modulation.

COMMERCIALISATION METRICS

Novelty

6/10

The paper provides a significant mechanistic insight into the temporal decoupling of auxin and GA signaling, but it characterizes a specific regulatory circuit rather than introducing a transformative new biological principle or platform technology.

Commercial Potential

4/10

The study provides fundamental mechanistic insights into plant hormonal signaling that could inform crop engineering, but it lacks a specific application, such as a targeted trait or a scalable delivery platform. While potentially useful for agronomic optimization, the direct path to a commercial product or licensable technology is currently indirect and academic.

Market Size

3/10

The research provides fundamental mechanistic insight into plant hormone crosstalk, but lacks a direct application to high-value agtech products or specific crop yield commercialization strategies. While potentially useful for long-term breeding, the abstract describes basic biological discovery rather than a defined, scalable market opportunity.

Tech Readiness

2/10

This is fundamental basic research focused on molecular signaling pathways in plant development. While it provides mechanistic insights, it lacks the applied validation, field trials, or prototype development required for higher technology readiness levels.

IP Strength

3/10

The paper describes fundamental biological mechanisms (auxin-GA crosstalk) that are likely natural phenomena, which are difficult to patent. While the discovery could inform new agrochemical targets, the intellectual property is limited to basic scientific knowledge rather than a specific, protectable composition or engineered tool.

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